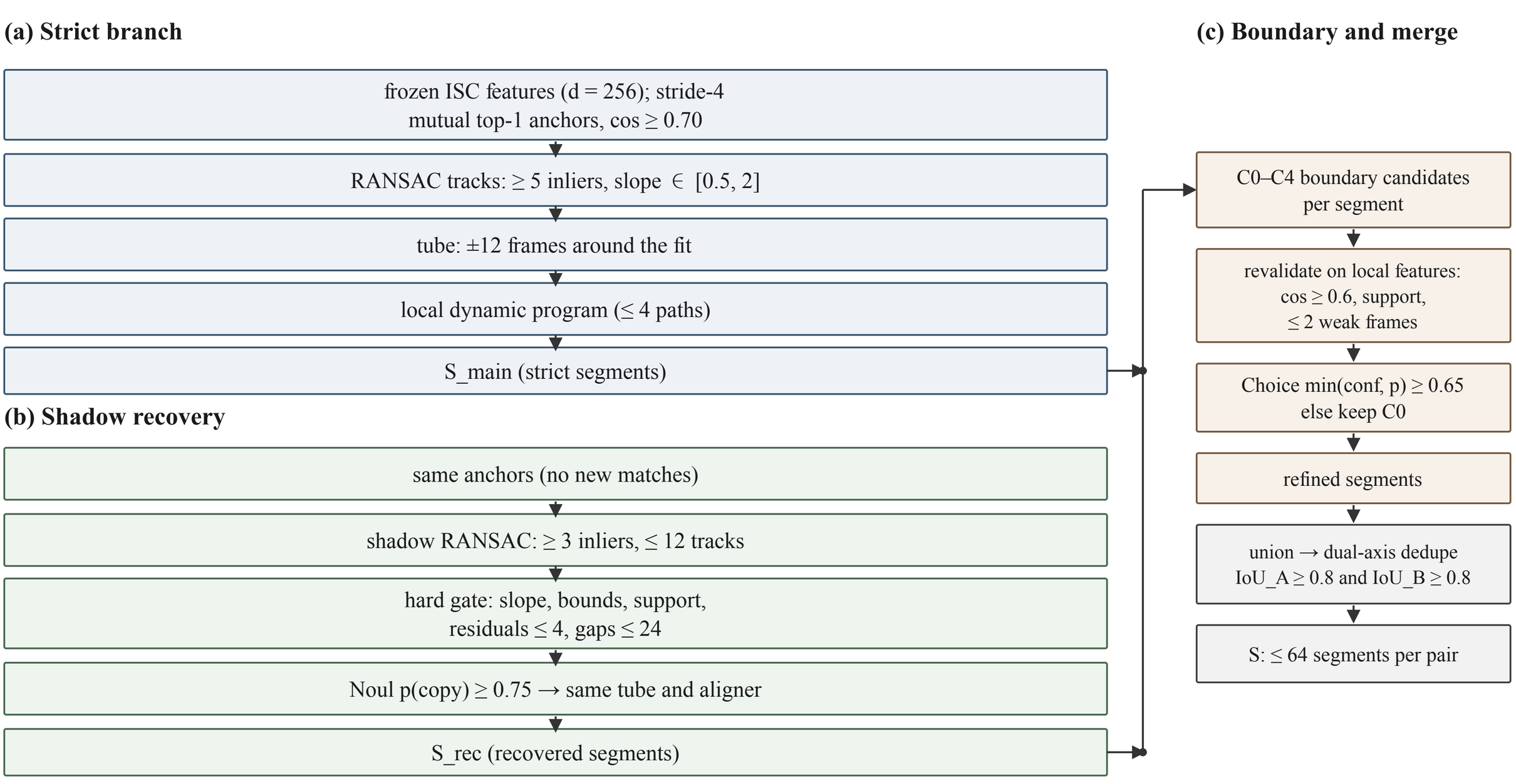

(a) Strict branch
frozen ISC features (d = 256); stride-4
mutual top-1 anchors, cos ≥ 0.70
RANSAC tracks: ≥ 5 inliers, slope ∈ [0.5, 2]
tube: ±12 frames around the fit
local dynamic program (≤ 4 paths)
S_main (strict segments)
(b) Shadow recovery
same anchors (no new matches)
shadow RANSAC: ≥ 3 inliers, ≤ 12 tracks
hard gate: slope, bounds, support,
residuals ≤ 4, gaps ≤ 24
Noul p(copy) ≥ 0.75 → same tube and aligner
S_rec (recovered segments)
(c) Boundary and merge
C0–C4 boundary candidates
per segment
revalidate on local features:
cos ≥ 0.6, support,
≤ 2 weak frames
Choice min(conf, p) ≥ 0.65
else keep C0
refined segments
union → dual-axis dedupe
IoU_A ≥ 0.8 and IoU_B ≥ 0.8
S: ≤ 64 segments per pair